



ABBOTSLEIGH

AUGUST 2008

YEAR 12

ASSESSMENT 4

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks – 120

- Attempt Questions 1-8.
- All questions are of equal value.
- Answer each question in a new booklet.

Outcomes assessed

HSC course

- E1** appreciates the creativity, power and usefulness of mathematics to solve a broad range of problems
- E2** chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings
- E3** uses the relationship between algebraic and geometric representations of complex numbers and of conic sections
- E4** uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials
- E5** uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces and resisted motion
- E6** combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions
- E7** uses the techniques of slicing and cylindrical shells to determine volumes
- E8** applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems
- E9** communicates abstract ideas and relationships using appropriate notation and logical argument

Harder applications of the Extension 1 Mathematics course are included in this course. Thus the Outcomes from the Extension 1 Mathematics course are included.

From the Extension 1 Mathematics Course

Preliminary course

- PE1** appreciates the role of mathematics in the solution of practical problems
- PE2** uses multi-step deductive reasoning in a variety of contexts
- PE3** solves problems involving inequalities, polynomials, circle geometry and parametric representations
- PE4** uses the parametric representation together with differentiation to identify geometric properties of parabolas
- PE5** determines derivatives that require the application of more than one rule of differentiation
- PE6** makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations

HSC course

- HE1** appreciates interrelationships between ideas drawn from different areas of mathematics
- HE2** uses inductive reasoning in the construction of proofs
- HE3** uses a variety of strategies to investigate mathematical models of situations involving projectiles, simple harmonic motion or exponential growth and decay
- HE4** uses the relationship between functions, inverse functions and their derivatives
- HE5** applies the chain rule to problems including those involving velocity and acceleration as functions of displacement
- HE6** determines integrals by reduction to a standard form through a given substitution
- HE7** evaluates mathematical solutions to problems and communicates them in an appropriate form

Total marks – 120

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

QUESTION 1 (15 marks)

Use a SEPARATE writing booklet.

(a) Using the table of standard integrals find $\int \frac{dx}{\sqrt{x^2 + 7}}$ **1**

(b) By completing the square find $\int \frac{dx}{\sqrt{4x - x^2}}$ **2**

(c) Find $\int \frac{1-2x}{\sqrt{1-x^2}} dx$, $|x| < 1$. **2**

(d) Find $\int \cos^3 x dx$. **2**

(e) (i) Use the substitution $x = \frac{2}{3} \sin \theta$ to prove that $\int_0^{\frac{2}{3}} \sqrt{4-9x^2} dx = \frac{\pi}{3}$. **3**

(ii) Hence or otherwise, find the area enclosed by the ellipse $\frac{9x^2}{4} + \frac{y^2}{4} = 1$. **2**

(f) Evaluate $\int_0^1 \tan^{-1} x dx$. **3**

QUESTION 2 (15 marks)
Start a new writing booklet.

(a) Given $z_1 = 3 - i$ and $z_2 = 2 + 5i$, express the following in the form $a + ib$ where a and b are real:

(i) $(\bar{z}_1)^2$ 2

(ii) $\frac{z_1}{z_2}$ 2

(iii) $|z_1 z_2|$ 2

(b) (i) Sketch the region $|z + 1 + i| \leq 1$. 2

(ii) Find the maximum and minimum values of $|z|$. 2

(c) (i) The complex number $z = x + iy$ is represented by the point P . If $\frac{z-1}{z-2i}$ is purely imaginary, show that the locus of P is the circle $x^2 - x + y^2 - 2y = 0$. 3

(ii) Sketch this locus showing all important features. 2

QUESTION 3 (15 marks)
Start a new writing booklet.

(a) Sketch on separate diagrams, the graphs of:

(i) $y = (x-1)^2(x+2)$ **1**

(ii) $y^2 = (x-1)^2(x+2)$ **2**

(iii) $y = \frac{1}{(x-1)^2(x+2)}$ **2**

(b) Sketch $y = \log_e(x+1)^2$ **2**

(c) Sketch the graph of the function $y = \frac{x^2 - x + 1}{(x-1)^2}$, clearly showing the coordinates of any points of intersection with the x and y axes, the coordinates of any turning points and the equations of any asymptotes. There is no need to investigate points of inflexion. **4**

(d) If α , β and γ are the roots of $x^3 + 2x^2 - 3x - 4 = 0$,

(i) Evaluate $\alpha^2 + \beta^2 + \gamma^2$. **2**

(ii) Form the equation whose roots are $\beta\gamma$, $\alpha\gamma$ and $\alpha\beta$. **2**

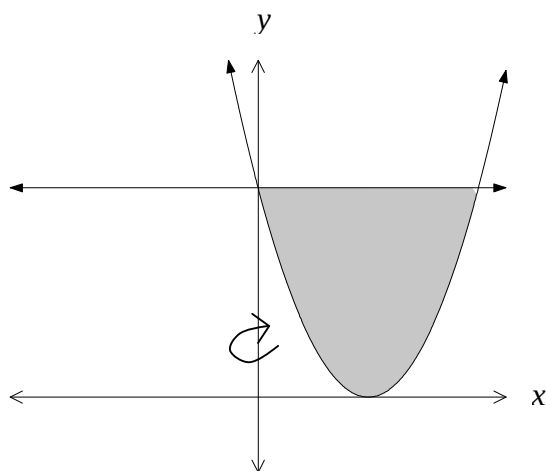
QUESTION 4 (15 marks)
Start a new writing booklet.

- (a) The foci of a hyperbola of eccentricity $\frac{13}{12}$ are the points $(\pm 13, 0)$.
- (i) Show that the equation of the hyperbola is $\frac{x^2}{144} - \frac{y^2}{25} = 1$. **2**
- (ii) Find the equation of the tangent to the hyperbola at the point $(12\sec\theta, 5\tan\theta)$. **3**
- (b) (i) Show that the condition for the line $y = mx + c$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is that $c^2 = a^2m^2 + b^2$. **2**
- (ii) Show that the pair of tangents drawn from the point $(3, 4)$ to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are at right angles to each other. **3**
- (c) (i) Verify that $\alpha = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$ is a root of $z^5 + z - 1 = 0$. **2**
- (ii) Find the monic cubic equation with real coefficients whose roots are also the roots of $z^5 + z - 1 = 0$ but do not include α . **3**

QUESTION 5 (15 marks)
Start a new writing booklet.

- (a) The base of a certain solid is a circle with radius 2. Each parallel cross-section of the solid is a square. Find the volume of the solid. **3**

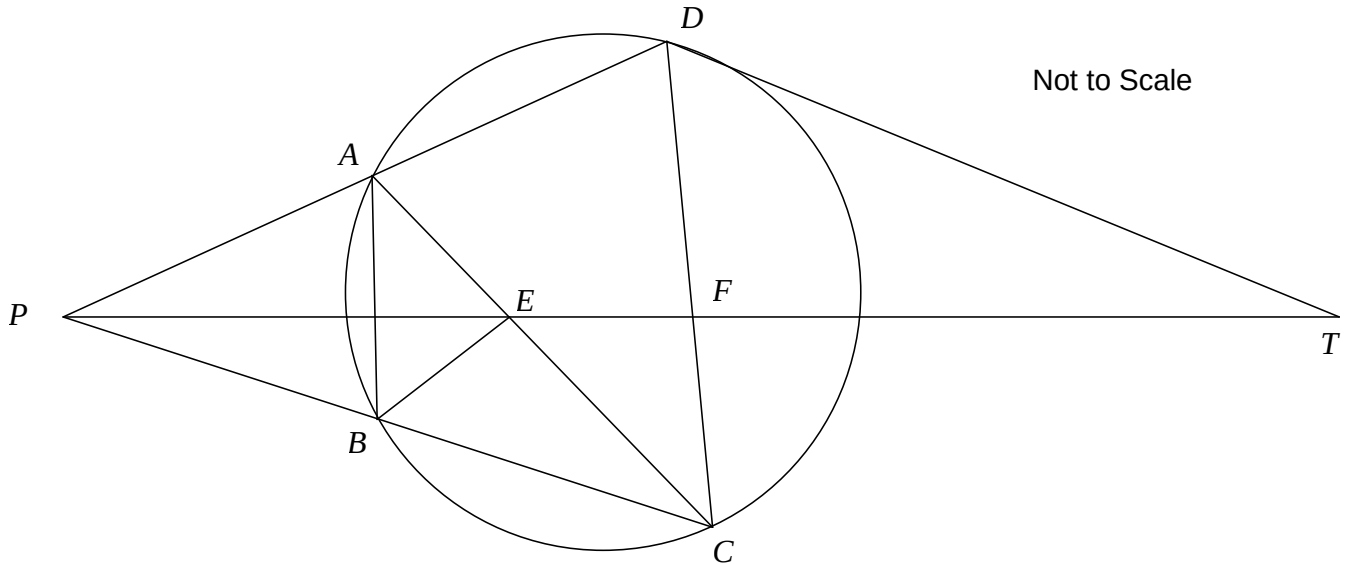
- (b) The area enclosed by the curve $y = (x-2)^2$ and the line $y = 4$ is rotated around the y -axis. Use the method of cylindrical shells to find the volume formed. **3**



- (c) (i) Show that the tangent to the rectangular hyperbola $xy = 4$ at the point $T\left(2t, \frac{2}{t}\right)$ has equation $x + t^2y = 4t$. **2**
- (ii) This tangent cuts the x -axis at the point Q . Find the coordinates of Q . **1**
- (iii) Show that the line through Q which is perpendicular to the tangent at T has equation $t^2x - y = 4t^3$. **1**
- (iv) This line through Q cuts the rectangular hyperbola at the points R and S . Show that the midpoint of RS has coordinates $M(2t, -2t^3)$. **3**
- (v) Find the equation of the locus of M as T moves on the rectangular hyperbola, stating any restrictions that may apply. **2**

QUESTION 6 (15 marks)
Start a new writing booklet.

- (a) $ABCD$ is a cyclic quadrilateral. DA produced and CB produced meet at P . T is a point on the tangent at D . PT cuts CA and CD at E and F respectively. $TF = TD$.



- (i) Copy the diagram and show that $AEFD$ is a cyclic quadrilateral. 3
- (ii) Show that $AEBP$ is a cyclic quadrilateral. 2
- (b) (i) If $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$ prove that $2nI_{n+1} = 2^{-n} + (2n-1)I_n$. 3
- (ii) Hence evaluate $\int_0^1 \frac{dx}{(1+x^2)^3}$. 2
- (c) (i) Use the substitution $x = a - y$ where a is a constant to prove that
- $$\int_0^a f(x)dx = \int_0^a f(a-x)dx$$
- 1
- (ii) Hence show that $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$ 4

QUESTION 7 (15 marks)

Start a new writing booklet.

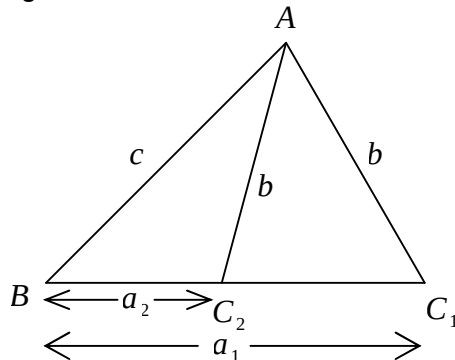
- (a) The functions $S(x)$ and $C(x)$ are defined by the formulae:

$$S(x) = \frac{1}{2}(e^x - e^{-x}) \text{ and } C(x) = \frac{1}{2}(e^x + e^{-x}).$$

- (i) Verify that $S'(x) = C(x)$. 1
- (ii) Show that $S(x)$ is an increasing function for all real values of x . 1
- (iii) Prove that $[C(x)]^2 = 1 + [S(x)]^2$. 1
- (iv) $S(x)$ has an inverse function $S^{-1}(x)$ for all values of x . Briefly justify this statement. 1
- (v) Let $y = S^{-1}(x)$. Prove $\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}}$. 2
- (vi) Hence, or otherwise, show that $S^{-1}(x) = \log_e \left(x + \sqrt{1+x^2} \right)$. 2

- (b) (i) Using the remainder theorem, or otherwise, show that $x - a - b - c$ is a factor of $P(x) = (x-a)(x-b)(x-c) - (b+c)(c+a)(a+b)$. 2
- (ii) Hence, or otherwise, solve the equation $(x-2)(x+3)(x+1) - 4 = 0$. 2

- (c) In $\triangle ABC$ the lengths b and c and $\angle B$ are given and have such values that two distinct triangles are possible as shown in the diagram below.



Show that $a_1 - a_2 = 2\sqrt{b^2 - c^2 \sin^2 B}$

3

QUESTION 8 (15 marks)
Start a new writing booklet.

Marks

- (a) A particle of mass 1 kg moves in a straight line before coming to rest. The resultant force acting on the particle directly opposes its motion and has magnitude $m(1+v)$ where v is its velocity. Initially the particle is at the origin and travelling with velocity Q where $Q > 0$
- (i) Show that v is related to the displacement x by the formula $x = Q - v + \log_e \left(\frac{1+v}{1+Q} \right)$. **3**
- (ii) Find an expression for v in terms of t . **2**
- (iii) Find an expression for x in terms of t . **1**
- (iv) Show that $Q = x + v + t$ **1**
- (v) Find the distance travelled and the time taken by the particle in coming to rest. **2**
- (b) (i) State why, for $x < 1$, the sum of n terms of the series $1 + x + x^2 + x^3 + \dots + x^{n-1}$ is $\frac{1-x^n}{1-x}$. **1**
- (ii) Show that $1 + 2x + 3x^2 + \dots + (n-1)x^{n-2} = \frac{(n-1)x^n - nx^{n-1} + 1}{(1-x)^2}$ **2**
- (iii) Hence find an expression for $1 + 1 + \frac{3}{4} + \frac{4}{8} + \dots + \frac{n-1}{2^{n-2}}$ and show that this sum is always less than 4. **3**

END OF PAPER

TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$



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Mathematics Extension 2

Solutions

Question 1

$$(a) = \ln(x + \sqrt{x^2 + 7}) + C$$

$$(b) = \int \frac{dx}{\sqrt{4 - (x - 2)^2}}$$

$$= \sin^{-1} \frac{x - 2}{2} + C$$

$$(c) = \int \frac{1}{\sqrt{1 - x^2}} - 2x(1 - x^2)^{-\frac{1}{2}} dx$$

$$= \sin^{-1} x - 2\sqrt{1 - x^2} + C$$

$$(d) = \int \cos x (1 - \sin^2 x) dx$$

$$= \sin x - \frac{1}{3} \sin^3 x + C$$

(e) i.

$$dx = \frac{2}{3} \cos \theta d\theta, \quad \begin{cases} x = 0, \sin \theta = 0 \\ x = \frac{2}{3}, \sin \theta = \frac{\pi}{2} \end{cases}$$

$$\therefore I = \int_0^{\frac{\pi}{2}} \frac{4}{3} \cos^2 \theta d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{2}{3} (\cos 2\theta + 1) d\theta$$

$$= \left[\frac{1}{3} \sin 2\theta + \frac{2}{3} \theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{3}$$

ii.

$$\text{Note that } \int_0^{\frac{2}{3}} \sqrt{4 - 9x^2} dx = \int_{\frac{2}{3}}^0 \sqrt{4 - 9x^2} dx$$

Now, the top part of the ellipse has the equation $y = \sqrt{4 - 9x^2}$

$$A_{top} = 2I = \frac{2\pi}{3}, \quad (\text{from part i})$$

Similarly, due to symmetry, the bottom part of the ellipse has the same area

$$A_{total} = \frac{4\pi}{3}$$

(f) Observing the graph of $y = \tan^{-1} x$ between $(0, 0)$ and $(1, \frac{\pi}{4})$

$$I = \frac{\pi}{4} - \int_0^{\frac{\pi}{4}} \tan y dy$$

$$= \frac{\pi}{4} + [\ln(\cos x)]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

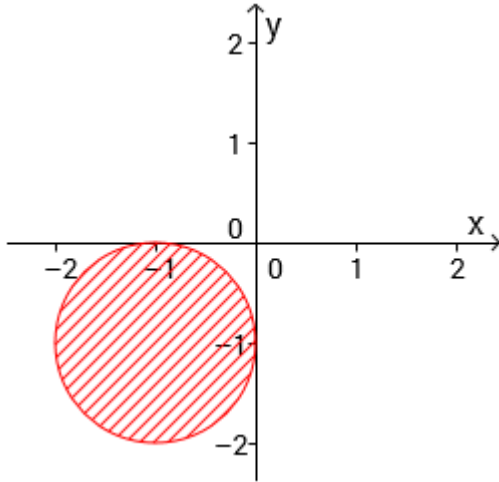
Question 2

(a) i. $(\bar{z}_1)^2 = 4 - 6i$

ii.
$$\frac{z_1}{z_2} = \frac{(3-i)(2-5i)}{29}$$
$$= \frac{1-17i}{29}$$

iii.
$$z_1 z_2 = 1 + 17i$$
$$|z_1 z_2| = \sqrt{290}$$

(b) i.



ii.

Distance of centre of locus from $(0, 0) = \sqrt{2}$

Minimum value of $|z| = \sqrt{2} - 1$, maximum value of $|z| = \sqrt{2} + 1$

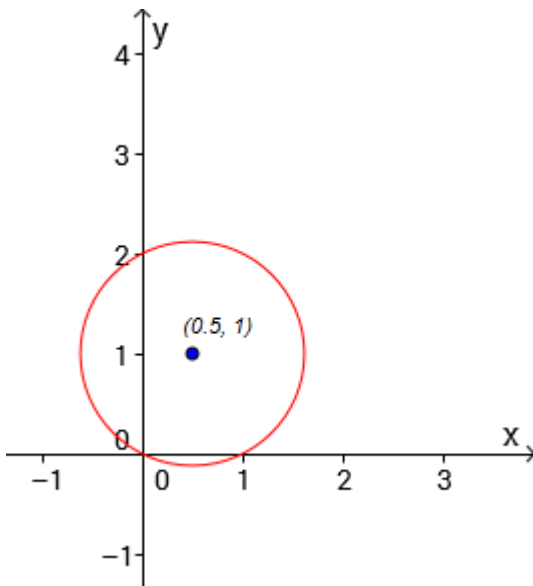
(c) i.

$$\frac{z-1}{z-2i} = \frac{x+iy-1}{x+iy-2i}$$
$$= \frac{(x+iy-1)(x-iy+2i)}{x^2+(y-2)^2}$$
$$= \frac{x^2+2ix+y^2-2y-x-iy-2i}{x^2+(y-2)^2}$$

Since the real part of $\frac{z-1}{z-2i}$ is 0 (i.e. the expression is purely imaginary)

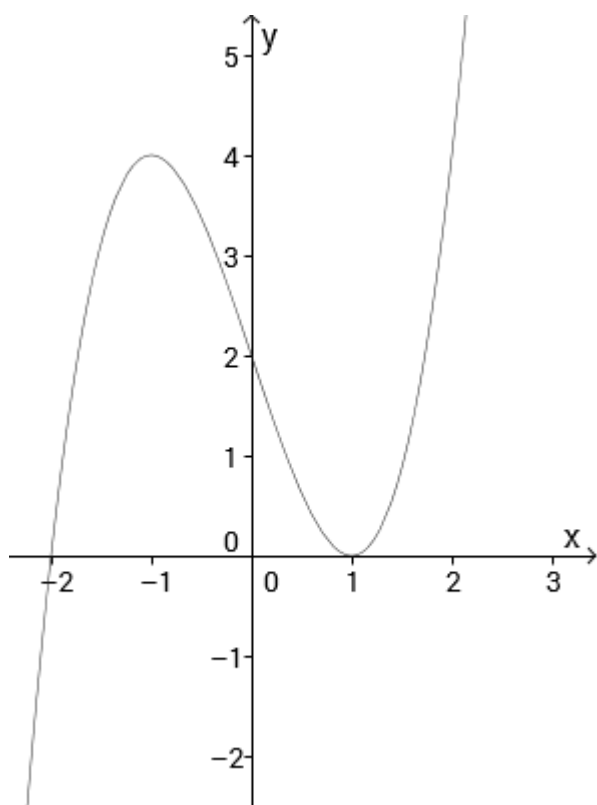
It follows that $x^2 - x + y^2 - 2y = 0$

ii.

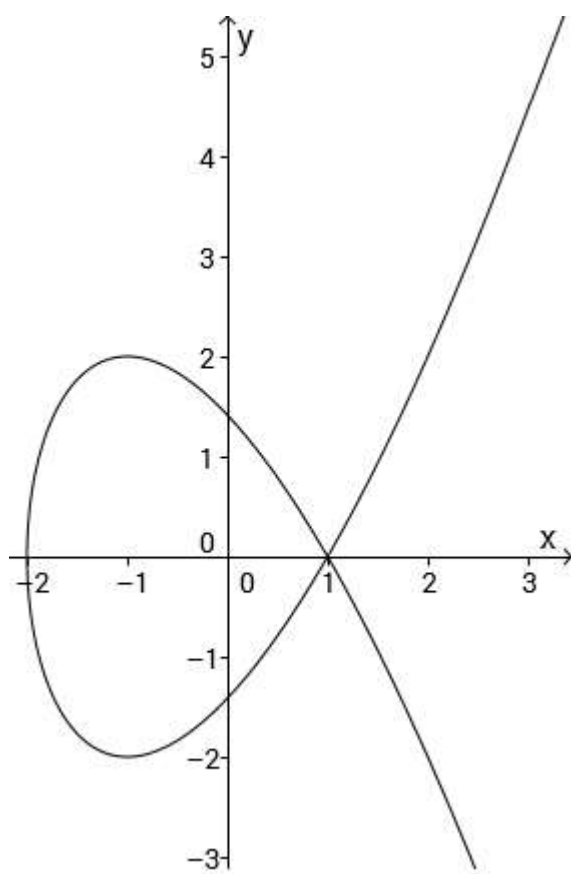


Question 3

(a) i.

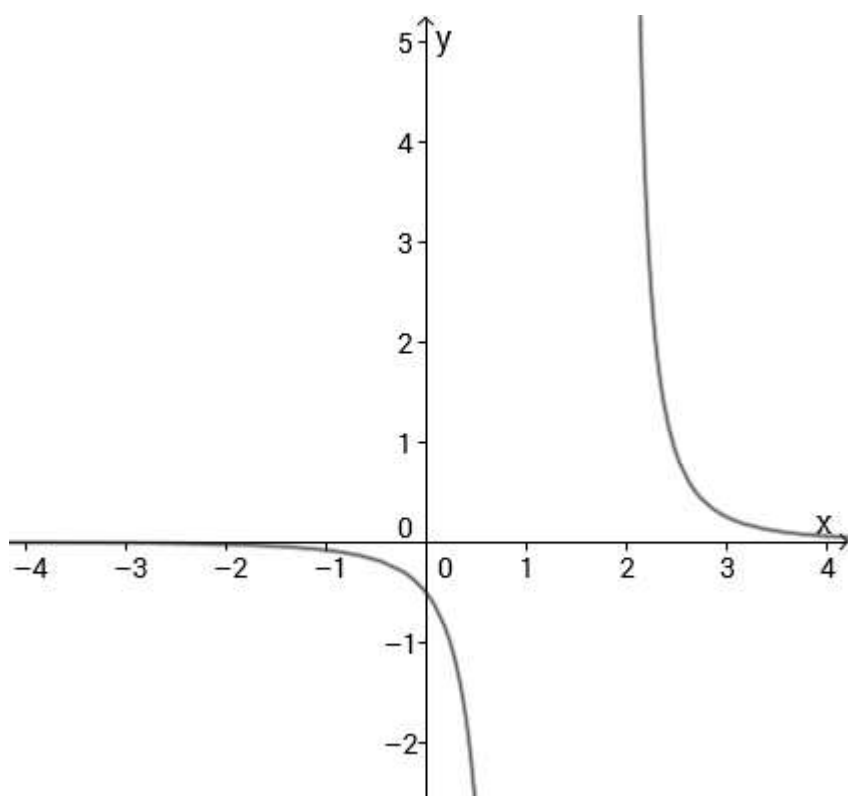


ii.

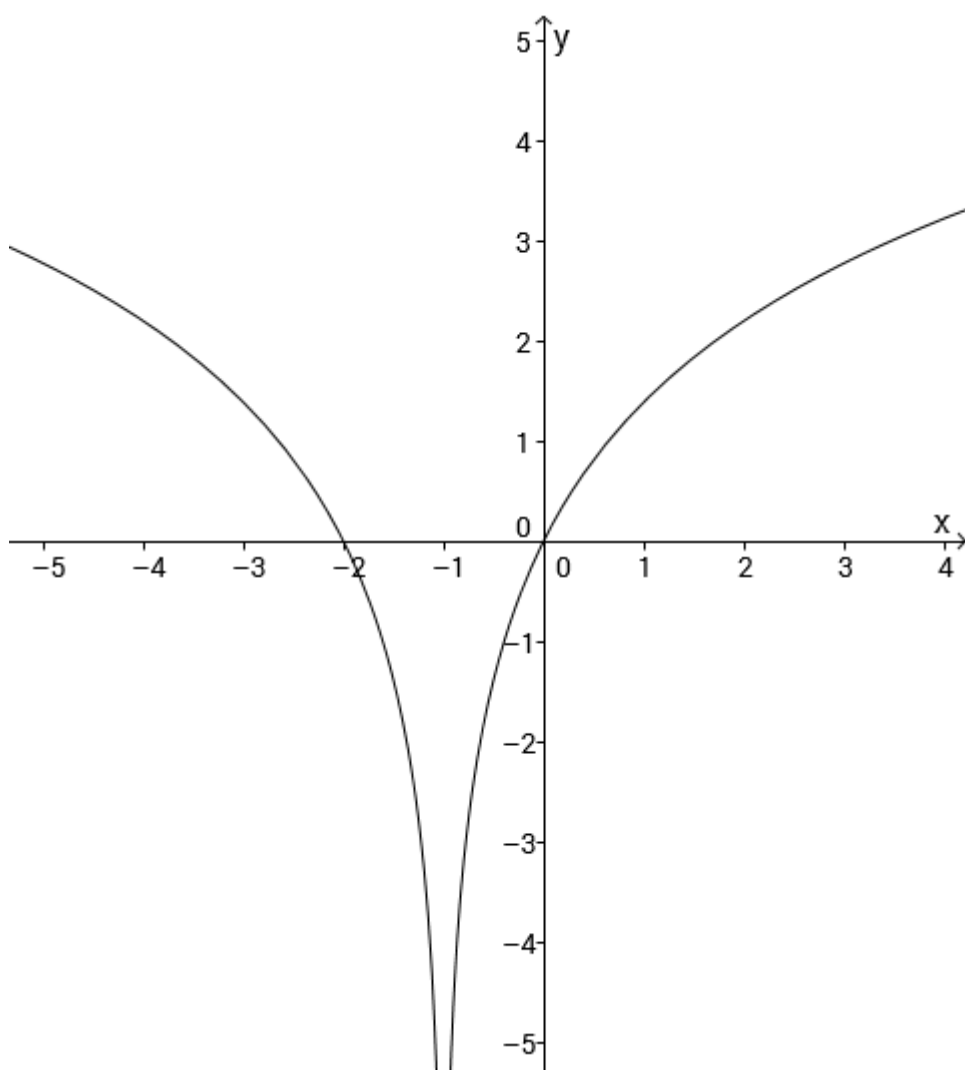


Question 3 (continued)

(a) iii.



(b)



Question 3 (continued)

(c) $y = x - 1 + \frac{x}{(x-1)^2}$

$$\frac{dy}{dx} = -\frac{x+1}{(x-1)^3}, \quad \frac{d^2y}{dx^2} = \frac{2(x+2)}{(x-1)^4}$$

At S.P.s, $\frac{dy}{dx} = 0$

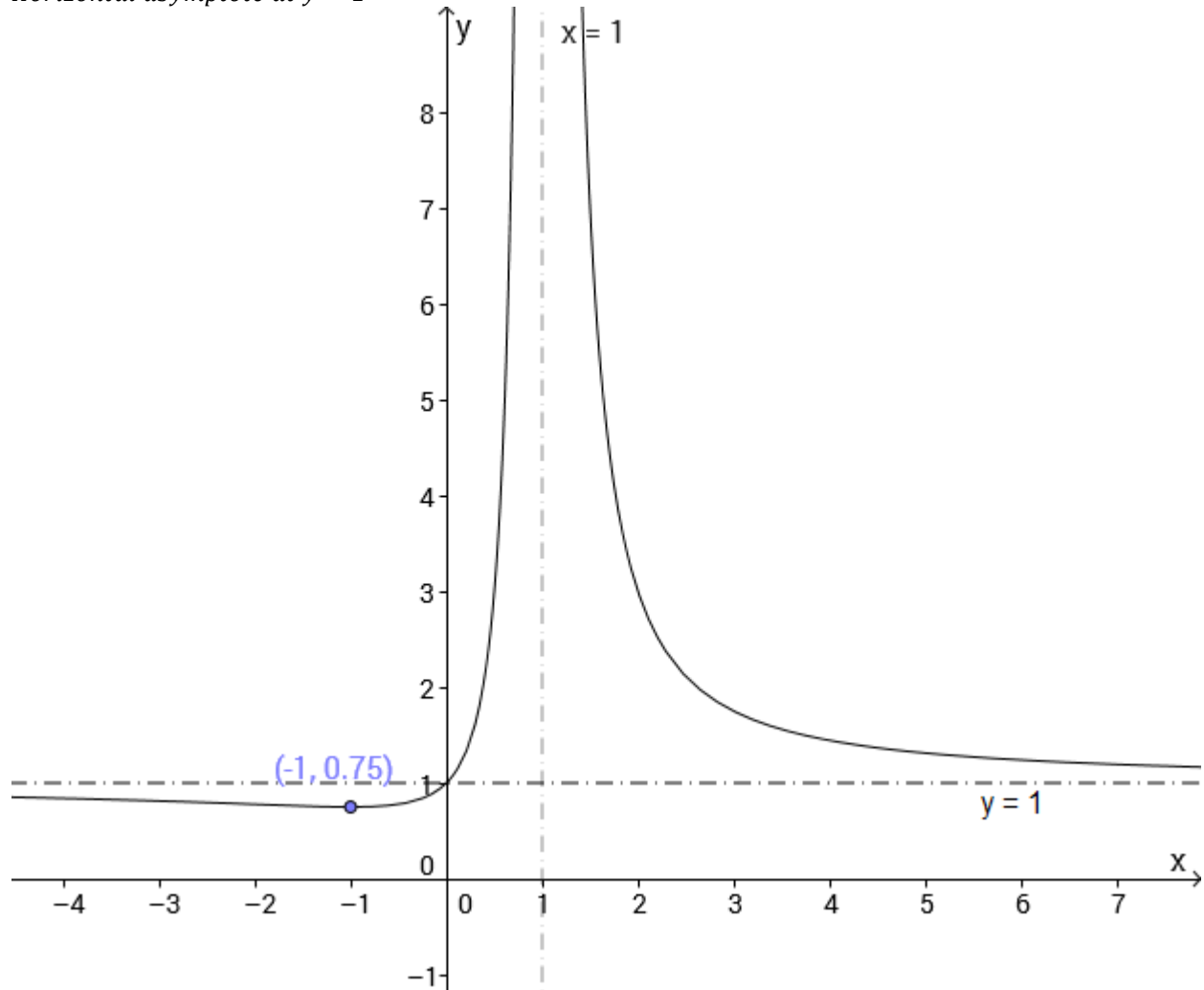
$$\begin{cases} x = -1 \\ y = 0.75 \end{cases}$$

$y'' > 0 \therefore \min T.P$

Vertical asymptote at $x = 1$

As $x \rightarrow \pm \infty$, $f(x) \rightarrow 1$

Horizontal asymptote at $y = 1$



(d) i.
 $\Sigma\alpha = -2, \quad \Sigma\alpha\beta = -3, \quad \alpha\beta\gamma = 4$
 $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2 \times \Sigma\alpha\beta$
 $= 4 + 6$
 $= 10$

ii.
 Let the roots be A, B and C
 $A + B + C = \alpha\beta + \beta\gamma + \gamma\alpha = -3$
 $AB + BC + AC = \alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha + \beta + \gamma) = 6$
 $ABC = (\alpha\beta\gamma)^2 = 16$

Therefore the equation required is $x^3 + 3x^2 + 6x - 16 = 0$

Question 4

(a) i.

$$e = \frac{13}{12}, \quad ae = \pm 13$$

$$a = \pm 12, \quad a^2 = 144$$

Using $b^2 = a^2(e^2 - 1)$, noting that foci lie on x axis

$$b^2 = 25$$

$$\text{Therefore equation is } \frac{x^2}{144} - \frac{y^2}{25} = 0$$

ii.

$$25x^2 - 144y^2 = 0$$

$$25x - 144y \frac{dy}{dx} = 0 \text{ (implicit differentiation and then dividing all terms by 2)}$$

$$\frac{dy}{dx} = \frac{25x}{144y}$$

$$\text{At } (12 \sec \theta, 5 \tan \theta), m_T = \frac{5 \sec \theta}{12 \tan \theta}$$

$$12y \tan \theta - 60 \tan^2 \theta = 5x \sec \theta - 60 \sec^2 \theta$$

$$5x \sec \theta - 12y \tan \theta - 60 = 0$$

(b) i.

Intersect $y = mx + c$ with ellipse gives

$$b^2x^2 + a^2m^2x^2 + 2a^2mcx + a^2c^2 - a^2b^2 = 0$$

$$(a^2m^2 + b^2)x^2 + (2a^2cm)x + a^2c^2 - a^2b^2 = 0$$

When $\Delta = 0$, $y = mx + c$ is a tangent (one solution)

$$4a^4c^2m^2 - 4(a^2m^2 + b^2)(a^2c^2 - a^2b^2) = 0$$

$$a^4c^2m^2 - a^4m^2c^2 + a^2b^2c^2 - a^2b^4 - a^4b^2m^2 = 0$$

$$a^2b^2c^2 - a^2b^4 - a^4b^2m^2 = 0$$

$$c^2 = a^2m^2 + b^2 \text{ (rearranging and dividing by } a^2b^2, a \neq 0, b \neq 0)$$

ii.

$$\text{Note that } c = 4 - 3m$$

For each of the tangents in the form $y = mx + c$

$$c^2 = a^2m^2 + b^2, \text{ from (a)}$$

$$c^2 = 16m^2 + 9$$

$$(4 - 3m)^2 = 16m^2 + 9$$

$$16 - 24m + 9m^2 = 16m^2 + 9$$

$$7m^2 + 24m - 7 = 0$$

$$\text{product of roots} = \alpha\beta = m_1m_2 = -\frac{7}{7} = -1, \quad \therefore \text{Tangents are perpendicular}$$

(c) i.

$$\alpha = \text{cis } \frac{\pi}{3}, \quad \alpha^5 = \beta = \text{cis } \frac{5\pi}{3} = \text{cis } -\frac{\pi}{3}$$

Test $P(\alpha)$

$$\alpha^5 + \alpha - 1 =$$

$$= 0, \quad \text{since } \alpha + \beta = 1$$

ii.

$$\text{Original equation is } z^5 + 0z^4 + 0z^3 + 0z^2 + z - 1 = 0$$

$$\text{The other root is } \beta = \text{cis } -\frac{\pi}{3}, \quad \alpha + \beta = 1, \alpha\beta = 1$$

Method 1:

$$\alpha \text{ and } \beta \text{ are roots to } z^2 - z + 1 = 0, \quad \text{Long division gives } z^3 + z^2 - 1 = 0$$

Method 2:

Let the other roots be A, B and C

$$A + B + C = -1, \quad AB + AC + BC = 0, \quad ABC = 1$$

$$\text{which gives the required equation } z^3 + z^2 - 1 = 0$$

Question 5

- (a) Take a typical strip of width Δx , height of $2\sqrt{4-x^2}$

$$\Delta V = 4(4-x^2)\Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{-2}^2 4(4-x^2)\Delta x$$

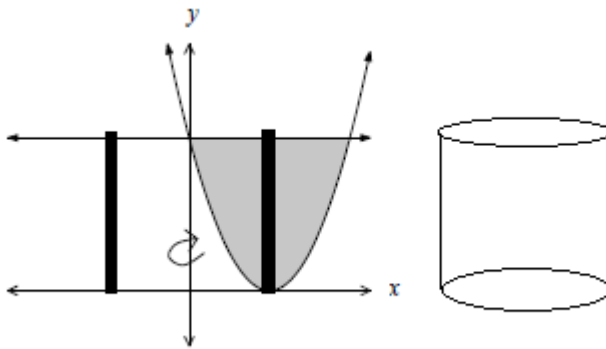
$$V = 4 \int_{-2}^2 4-x^2 dx$$

$$= 8 \int_0^2 4-x^2 dx$$

$$= 8 \left[4x - \frac{1}{3}x^3 \right]_0^2$$

$$= \frac{128}{3} \text{ cubic units.}$$

- (b)



Radius of typical shell = Δx

Height of typical shell = $(4-y) = 4 - (x-2)^2 = 4 - x^2 + 4x$

Circumference of typical shell = $2\pi x$

$$\Delta V = 2\pi x(4x - x^2)\Delta x$$

$$V = 2\pi \times \lim_{\Delta x \rightarrow 0} \sum_0^4 (4x^2 - x^3)\Delta x$$

$$V = 2\pi \int_0^4 4x^2 - x^3 dx$$

$$= 2\pi \left[\frac{4}{3}x^3 - \frac{1}{4}x^4 \right]_0^4$$

$$= \frac{128}{3} \pi \text{ cubic units.}$$

- (c) i.

$$y = \frac{4}{x}$$

$$\frac{dy}{dx} = -\frac{4}{x^2}$$

$$m_T \text{ (at } x = 2t) = -\frac{1}{t^2}$$

$$y - \frac{2}{t} = -\frac{1}{t^2}(x - 2t)$$

$$t^2 y - 2t = -x + 2t$$

$$x + t^2 y = 4t$$

- ii.

$$\text{At Q, } y = 0$$

$$\therefore \text{Q is } (4t, 0)$$

- iii.

$$m_{QS} = t^2$$

$$y = t^2(x - 4t)$$

$$t^2 x = 4t^3 + y, \quad (*)$$

- iv.

Intersect $xy = 4$ with $(*)$

$$t^2 x(x - 4t) = 4$$

$$t^2 x^2 - 4t^3 x - 4 = 0$$

$$\text{Sum of roots} = x_1 + x_2 = 4t$$

$$x_M = \frac{x_1 + x_2}{2} = 2t$$

Similarly

$$y(4t^3 + y) = 4t^2$$

$$y^2 + 4t^3 y - 4t^2 = 0$$

$$\text{Sum of roots} = y_1 + y_2 = -4y^3$$

$$y_M = \frac{y_1 + y_2}{2} = -2t^3$$

- v.

$$x = 2t$$

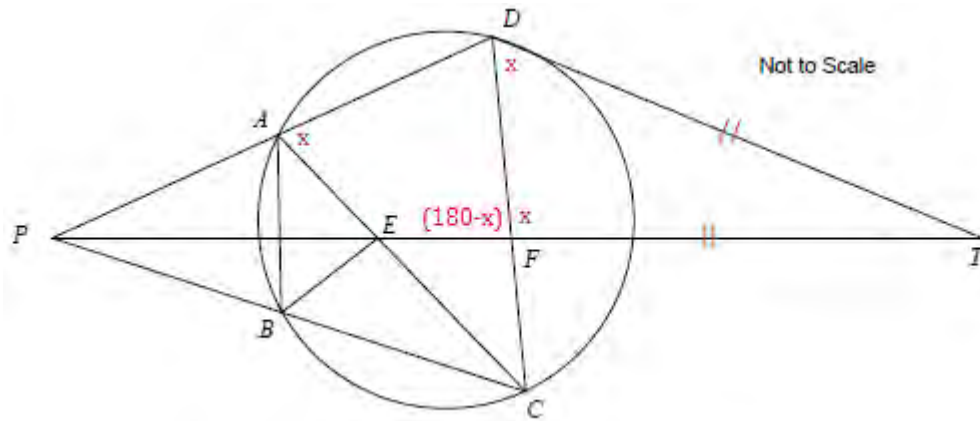
$$y = -2t^3$$

$$x^3 = 8t^3$$

$$\therefore \text{Locus of M is } x^3 = -4y$$

Question 6

(a) i.



Let $\angle CAD = x$

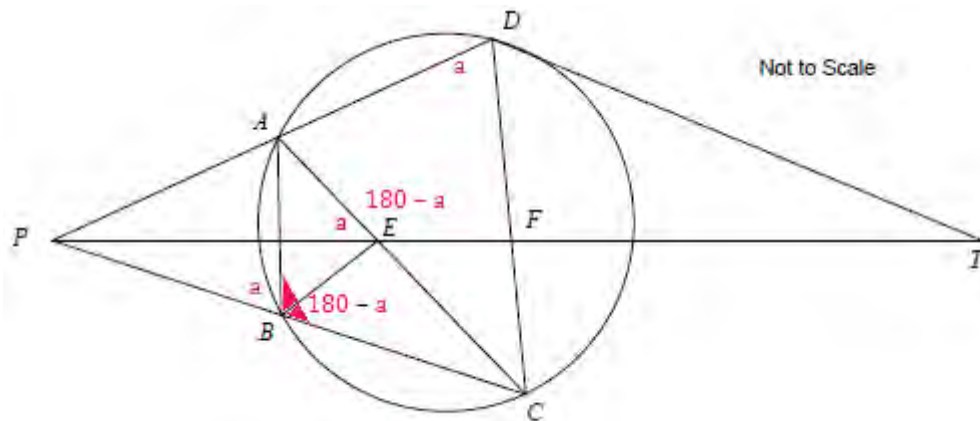
$\angle CDT = x$, (angle in the alternate segment theorem)

$\angle DFT = x$, (angles opposite equal sides of isosceles triangle equal)

$\angle EFD = 180 - x$, (angles in a straight line)

$\therefore AEF D$ is a cyclic quadrilateral, (opposite angles supplementary)

ii.



Let $\angle PEA = a$

$\angle AEF = 180 - a$, (angles on a straight line)

$\angle ADF = a$, (opposite angles in cyclic quadrilateral are supplementary & part i)

$\angle ABC = 180 - a$, (opposite angles in cyclic quadrilateral are supplementary)

$\angle ADF = a$, (angles on a straight line)

$\therefore AEBP$ is a cyclic quadrilateral, (angles standing on the same arc AP are equal)

(b) i.

$$I_n = \int_0^1 (1+x^2)^{-n} \frac{d}{dx}(x) dx$$

$$I_n = [x(1+x^2)^{-(n)}]_0^1 + 2n \int_0^1 x^2(1+x^2)^{-(n+1)} dx$$

$$I_n = 2^{-n} + 2n \int_0^1 (1+x^2)(1+x^2)^{-(n+1)} dx - 2n I_{n+1}$$

$$2n I_{n+1} = 2^{-n} + (2n-1) I_n$$

$$I_{n+1} = \frac{1}{n(2)^{(n+1)}} + \frac{2n-1}{2n} I_n$$

ii.

$$I_1 = \int_0^1 (1+x^2)^{-1} dx = [\tan^{-1} x]_0^1 = \frac{\pi}{4}$$

$$I_2 = \frac{1}{4} + \frac{\pi}{8}$$

$$I_3 = \frac{1}{16} + \frac{3}{4} \left(\frac{1}{4} + \frac{\pi}{8} \right) = \frac{1}{4} + \frac{3\pi}{32}$$

Question 6 (continued)

(c) i.

$$I = \int_0^a f(x) dx$$

$$\text{Let } x = a - y$$

$$dx = -dy$$

$$\begin{cases} x = a, y = 0 \\ x = 0, y = a \end{cases}$$

$$I = \int_0^a f(a - y) - dy$$

$$= \int_0^a f(a - y) dy$$

$$= \int_0^a f(a - x) dx, \quad \text{changing variable to } x$$

ii.

$$I = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$$

$$\therefore I = \int_0^\pi \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$$

$$I = \int_0^\pi \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$$

$$2I = \pi \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx$$

$$\text{Let } u = \cos x$$

$$\begin{cases} x = \pi, u = -1 \\ x = 0, u = 1 \end{cases}$$

$$du = -\sin x dx$$

$$2I = \pi \int_1^{-1} -\frac{1}{1 + u^2} du$$

$$2I = \pi \int_{-1}^1 \frac{1}{1 + u^2} du$$

$$2I = \pi [\tan^{-1} u]_{-1}^1$$

$$2I = \pi \left(\frac{\pi}{4} + \frac{\pi}{4} \right)$$

$$\therefore I = \frac{\pi^2}{4}$$

Question 7

(a) i.

$$S'(x) = \frac{1}{2}(e^x + e^{-x})$$

$$= C(x)$$

ii.

$$C(x) > 0 \forall x$$

$$S'(x) \geq 0 \forall x$$

iii.

$$LHS = \left(\frac{1}{2}(e^x + e^{-x}) \right)^2$$

$$= \frac{1}{4}(e^{2x} + e^{-2x}) + \frac{1}{2}$$

$$= 1 - \frac{1}{2} + \frac{1}{4}(e^{2x} + e^{-2x})$$

$$= 1 + \left(\frac{1}{2}(e^x - e^{-x}) \right)^2$$

$$= RHS$$

iv.

For every x value there is a unique y value.

v.

$$x = \frac{1}{2}(e^y - e^{-y})$$

$$\frac{dx}{dy} = \frac{1}{2}(e^y + e^{-y}), \quad \text{using part i.}$$

$$= C(y)$$

$$= \sqrt{1 + \frac{1}{2}(e^{-y} - e^{-y})}, \quad \text{using part iii.}$$

$$= \sqrt{1 + x^2}, \quad \text{from line 1}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 + x^2}}$$

vi.

$(0, 0)$ lies on $S(x)$

$\therefore$ by reflective property of inverse $(0, 0)$ lies on $S^{-1}(x)$

$$y = \int \frac{dx}{\sqrt{1 + x^2}}$$

$$y = \log_e \left(x + \sqrt{1 + x^2} \right) + C$$

Substituting the point $(0, 0)$ gives $C = 0$

$$\therefore y = S^{-1}(x) = \log_e \left(x + \sqrt{1 + x^2} \right)$$

(b) i.

If $(x - a - b - c)$ is a factor of $P(x)$ then $P(a + b + c) = 0$

$$P(a + b + c) = (a + b + c - a)(a + b + c - b)(a + b + c - c) - (b + c)(c + a)(a + b)$$

$$= (b + c)(c + a)(a + b) - (b + c)(c + a)(a + b)$$

$$= 0$$

ii.

$$P(x) = (x - a)(x - b)(x - c) - (a + b)(c + a)(b + c)$$

$$\text{Let } a = 2, \quad b = -3, \quad c = -1$$

$$P(x) = (x - 2)(x + 3)(x + 1) - 4$$

$\therefore$ from part i, $(x + 2)$ is a factor.

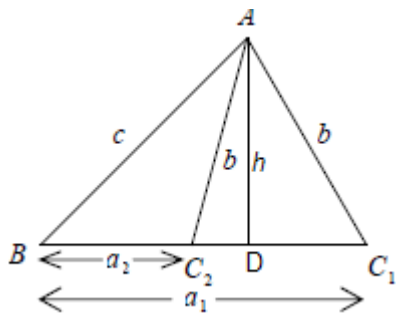
$$P(x) = x^3 + 2x^2 - 5x - 10$$

Long division gives $x^2 - 5$

Solutions to $P(x) = 0$

$$\text{are } x = -2, \quad x = \sqrt{5}, \quad x = -\sqrt{5}$$

(c)



Produce perpendicular from A

$$h = c \sin B$$

$$C_1 C_2 = 2\sqrt{b^2 - h^2}, \text{ noting that } C_2 D = C_1 D$$

$$\therefore a_1 - a_2 = 2\sqrt{b^2 - c^2 \sin^2 B}$$

Question 8

(a) i.

$$F = -m(1 + v)$$

$$a = -(1 + v)$$

$$v \frac{dv}{dx} = -(1 + v)$$

$$\frac{dv}{dx} = -\frac{1 + v}{v}$$

$$\frac{dx}{dv} = \frac{1}{1 + v} - \frac{1 + v}{1 + v}$$

$$x = \ln(1 + v) - v + C$$

$$\text{At } x = 0, \quad v = Q$$

$$C - Q + \ln(1 + Q) = 0$$

$$C = Q - \ln(1 + Q)$$

$$\therefore x = \ln(1 + v) - \ln(1 + Q) + Q - v$$

$$x = \ln\left(\frac{1 + v}{1 + Q}\right) + Q - v$$

ii.

$$a = -(1 + v)$$

$$\frac{dv}{dt} = -(1 + v)$$

$$\frac{dt}{dv} = -\frac{1}{(v + 1)}$$

$$t = -\ln(1 + v) + C$$

$$\text{At } t = 0, \quad v = Q$$

$$C = \ln(1 + Q)$$

$$t = \ln\left(\frac{1 + Q}{1 + v}\right)$$

iii.

$$e^{-t} = \frac{1 + v}{1 + Q}$$

$$v = (Q + 1)e^{-t} - 1 (*)$$

$$\frac{dx}{dt} = (Q + 1)e^{-t} - 1$$

$$x = -t - (Q + 1)e^{-t} + C$$

$$\text{At } t = 0, \quad x = 0$$

$$C = (Q + 1)$$

$$x = Q + 1 - t - (Q + 1)e^{-t}$$

iv.

$$Q = x + t + (Q + 1)e^{-1} - 1$$

$$Q = x + v + t, \quad \text{using } (*)$$

v.

Find when $v = 0$

$$x = Q - \ln(1 + Q)$$

$$t = \ln(1 + Q)$$

(b) i.

Geometric series for $r < 1$ is $\frac{a(1 - r^n)}{1 - r}$

In this series, $a = 1, r = 1$

which gives the sum as

$$\frac{1 - x^n}{1 - x}$$

ii.

$$1 + x + x^2 + \dots + x^{n-1} = \frac{(1 - x^n)}{1 - x}$$

Differentiate both sides

$$1 + 2x + 3x^2 + \dots + (n - 1)x^{n-2} = \frac{n(x - 1)x^{n-1} + 1 - x^n}{(1 - x)^2}$$

$$1 + 2x + 3x^2 + \dots + (n - 1)x^{n-2} = \frac{(n - 1)x^n - nx^{n-1} + 1}{(1 - x)^2}$$

iii.

$$\text{Let } x = \frac{1}{2}$$

$$LHS = 1 + 1 + \frac{3}{4} + \frac{(n - 1)}{2^{n-2}}$$

$$= 4 \left(-\left(\frac{1}{2}\right)^{n+1} - n\left(\frac{1}{2}\right)^{n-1} + 1 \right), \quad n > 0$$

$$= 4 - (2^{1-n})(4n + 1)$$

$$\leq 4 \forall n > 0$$